

Formulaire : trigonométrie circulaire

1 Formules fondamentales

$$\sin^2(x) + \cos^2(x) = 1$$

$$1 + \tan^2(x) = \frac{1}{\cos^2(x)}$$

$$1 + \cotan^2(x) = \frac{1}{\sin^2(x)}$$

2 Formules de symétrie

$$\sin(\pi + x) = -\sin(x)$$

$$\cos(\pi + x) = -\cos(x)$$

$$\tan(\pi + x) = \tan(x)$$

$$\sin(\pi - x) = \sin(x)$$

$$\cos(\pi - x) = -\cos(x)$$

$$\tan(\pi - x) = -\tan(x)$$

$$\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$$

$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$$

$$\cos\left(\frac{\pi}{2} + x\right) = -\sin(x)$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$$

$$\tan\left(\frac{\pi}{2} + x\right) = -\frac{1}{\tan(x)}$$

$$\tan\left(\frac{\pi}{2} - x\right) = \frac{1}{\tan(x)} = \cotan(x)$$

3 Transformation d'addition

$$\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$$

$$\tan(a + b) = \frac{\tan(a) + \tan(b)}{1 - \tan(a)\tan(b)}$$

$$\sin(a - b) = \sin(a)\cos(b) - \cos(a)\sin(b)$$

$$\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$$

$$\tan(a - b) = \frac{\tan(a) - \tan(b)}{1 + \tan(a)\tan(b)}$$

$$\cos(a - b) = \cos(a)\cos(b) + \sin(a)\sin(b)$$

4 Formules de duplication et de l'angle triple

$$\sin(2x) = 2\sin(x)\cos(x)$$

$$\tan(2x) = \frac{2\tan(x)}{1 - \tan^2(x)}$$

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\sin(3x) = 3\sin(x) - 4\sin^3(x)$$

$$\tan(3x) = \frac{3\tan(x) - \tan^3(x)}{1 - 3\tan^2(x)}$$

$$\cos(3x) = 4\cos^3(x) - 3\cos(x)$$

5 Formules de linéarisation

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

$$\sin^3(x) = \frac{-\sin(3x) + 3\sin(x)}{4}$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

$$\cos^3(x) = \frac{\cos(3x) + 3\cos(x)}{4}$$

$$\tan^2(x) = \frac{1 - \cos(2x)}{1 + \cos(2x)}$$

$$\tan^3(x) = \frac{-\sin(3x) + 3\sin(x)}{\cos(3x) + 3\cos(x)}$$

6 Transformation de produits en sommes

$$\sin(a) \sin(b) = \frac{1}{2} [\cos(a - b) - \cos(a + b)]$$

$$\cos(a) \cos(b) = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$$

$$\sin(a) \cos(b) = \frac{1}{2} [\sin(a + b) - \sin(b - a)]$$

7 Transformation de sommes en produits

$$\sin(a) - \sin(b) = 2 \sin\left(\frac{a - b}{2}\right) \cos\left(\frac{a + b}{2}\right) \quad \cos(a) - \cos(b) = -2 \sin\left(\frac{a + b}{2}\right) \sin\left(\frac{a - b}{2}\right)$$

$$\sin(a) + \sin(b) = 2 \sin\left(\frac{a + b}{2}\right) \cos\left(\frac{a - b}{2}\right) \quad \cos(a) + \cos(b) = 2 \cos\left(\frac{a + b}{2}\right) \cos\left(\frac{a - b}{2}\right)$$

8 Tangente de l'angle moitié : changement de variable

$$t = \tan\left(\frac{x}{2}\right) \quad \sin(x) = \frac{2t}{1 + t^2} \quad \cos(x) = \frac{1 - t^2}{1 + t^2} \quad \tan(x) = \frac{2t}{1 - t^2} \quad dx = \frac{2}{1 + t^2} dt$$

9 Formules en fonction de l'angle moitié

$$1 + \cos(x) = 2 \cos^2\left(\frac{x}{2}\right) \quad 1 - \cos(x) = 2 \sin^2\left(\frac{x}{2}\right) \quad \sin(x) = 2 \cos\left(\frac{x}{2}\right) \sin\left(\frac{x}{2}\right)$$